

ML-Based Automata Simplification for Symbolic Accelerators

Tiffany Yu, Rye Stahle-Smith, Dr. Rasha Karakchi
tyu@email.sc.edu, RYES@email.sc.edu, karakchi@cec.sc.edu

Magellan Scholar
McNair Junior Fellowship

Graduation with
Leadership Distinction in Research

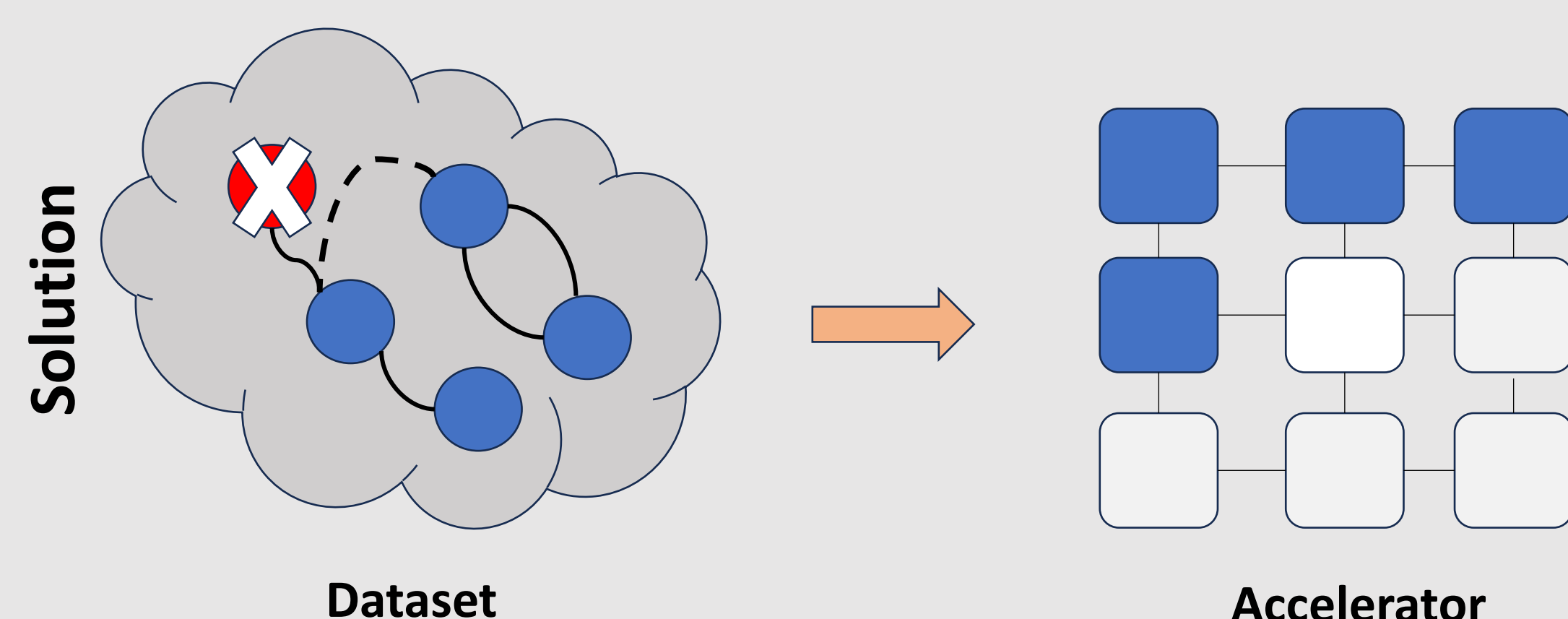
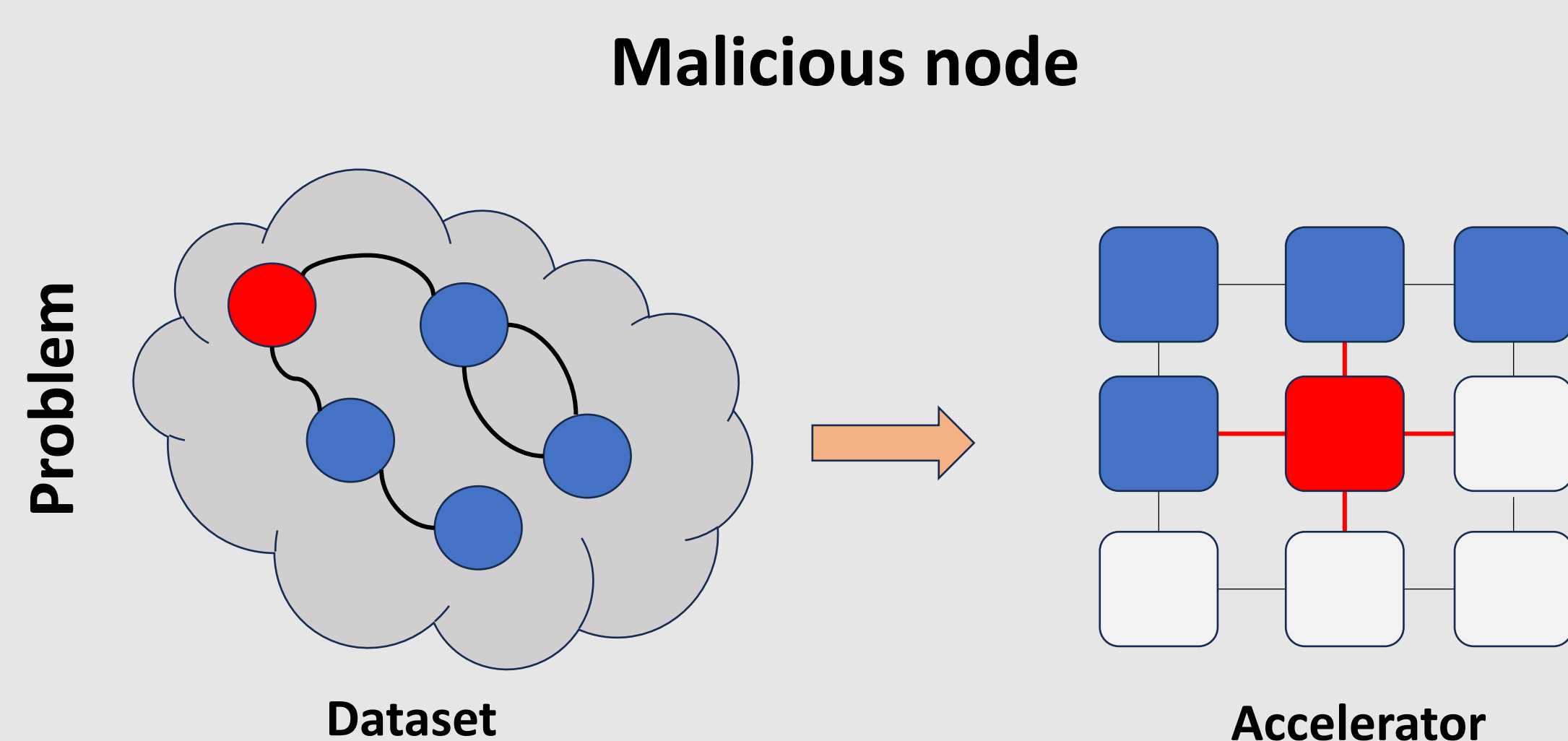
Motivation

- Modern computational hardware, including CPUs, GPUs, and AI accelerators, relies on efficient memory access for performance and security.
- However, unnecessary memory accesses introduce vulnerabilities such as data leakage, side-channel attacks, and increased power consumption.
- Addressing these inefficiencies is crucial for ensuring both security and optimal resource utilization.

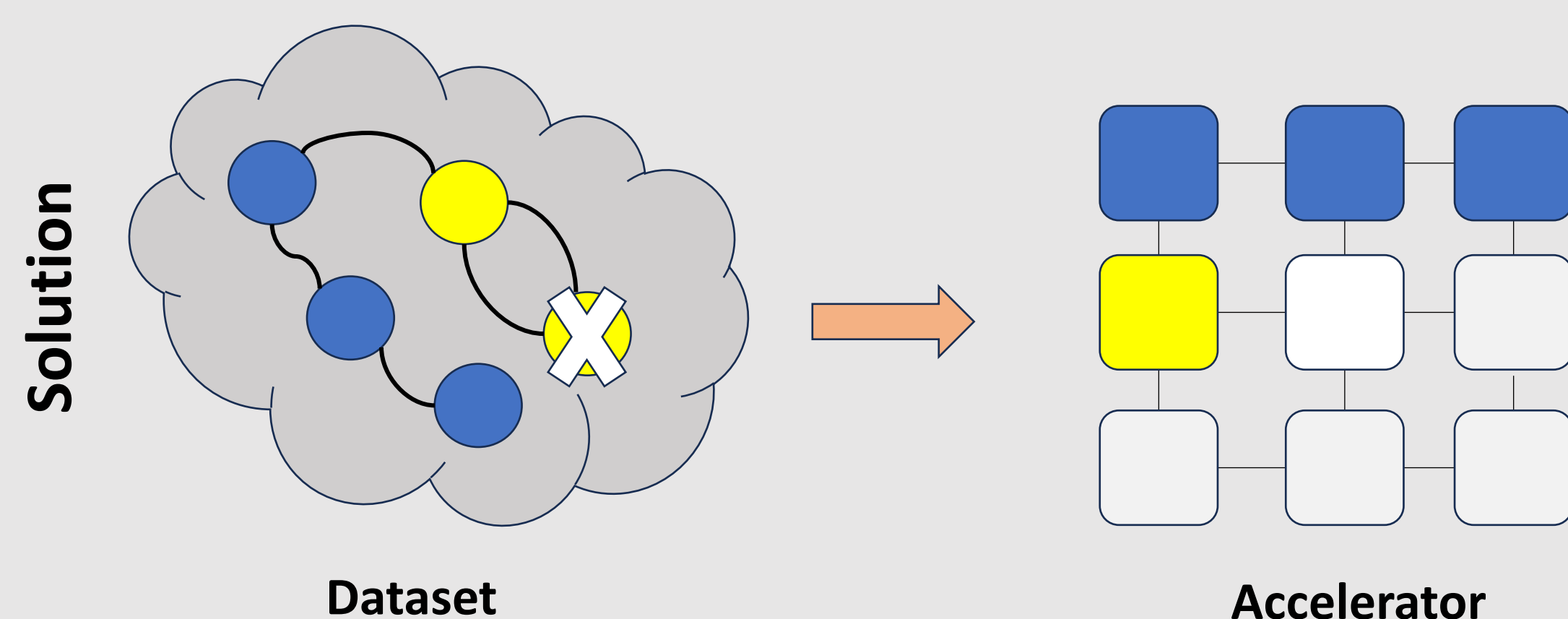
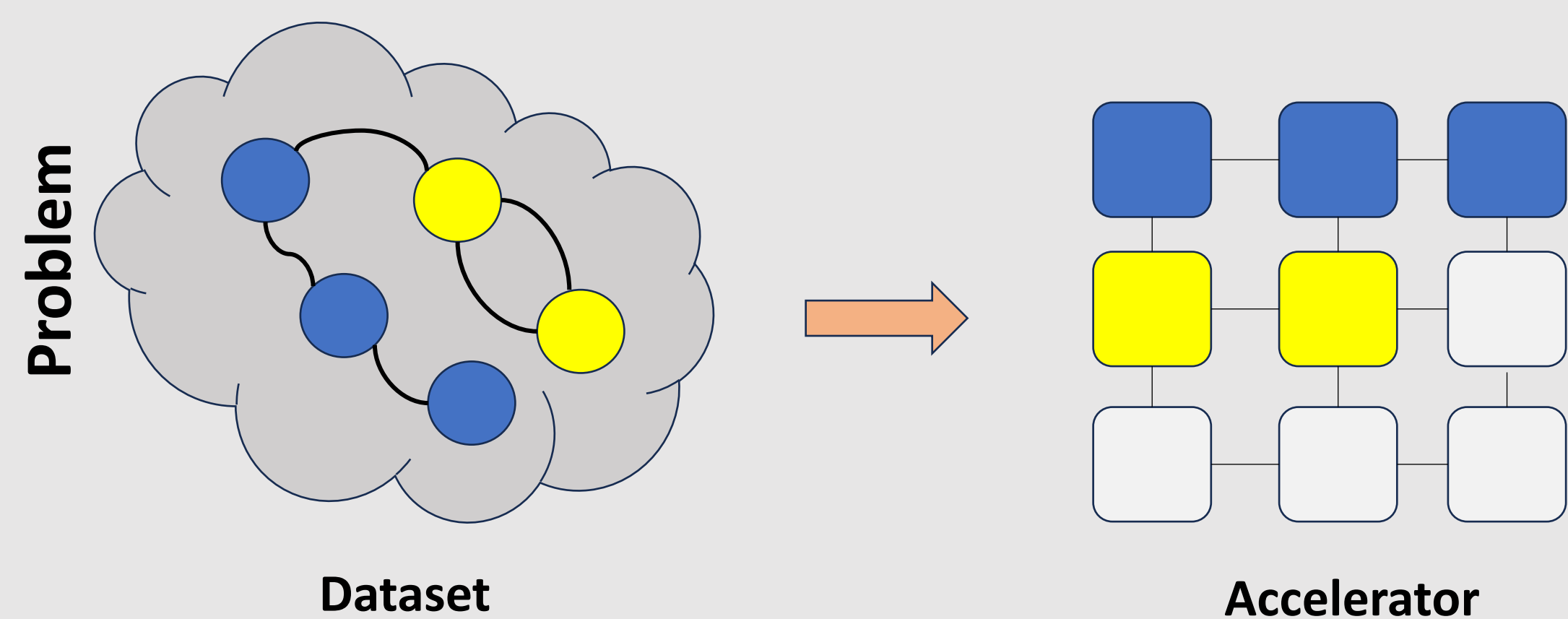
Problem

- Unnecessary or redundant memory accesses create security risks and degrade system efficiency.
- Data leakage:** Unused or poorly managed memory pathways provide opportunities for malicious code insertion and unauthorized data extraction
- Performance and Energy Overhead:** Frequent redundant memory accesses lead to higher latency, increased power consumption, and inefficient memory utilization, impacting computational workloads

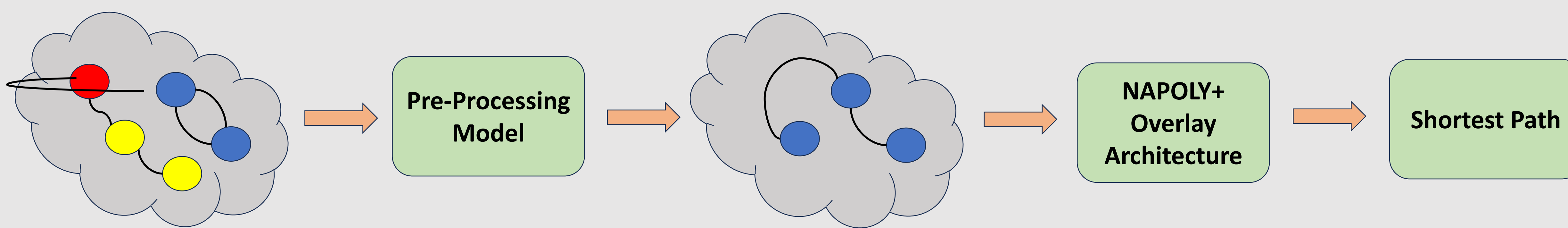
Explanation



Unnecessary node

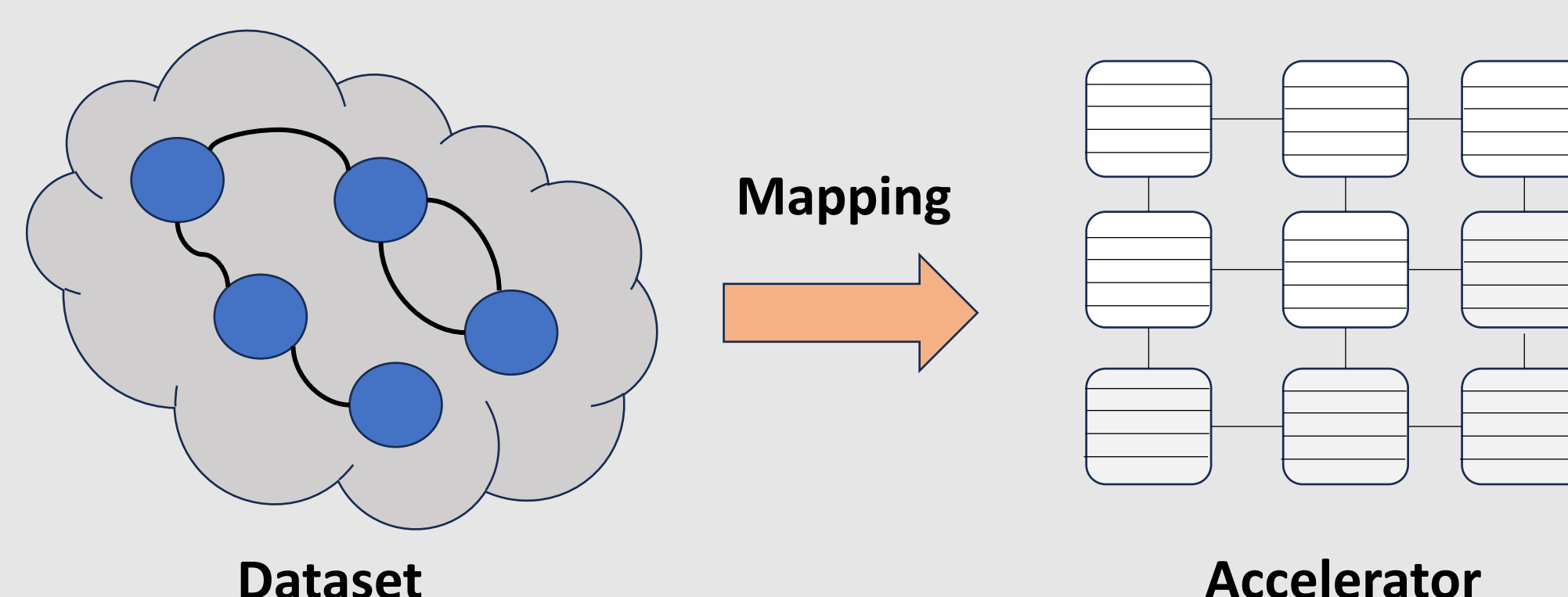


Framework



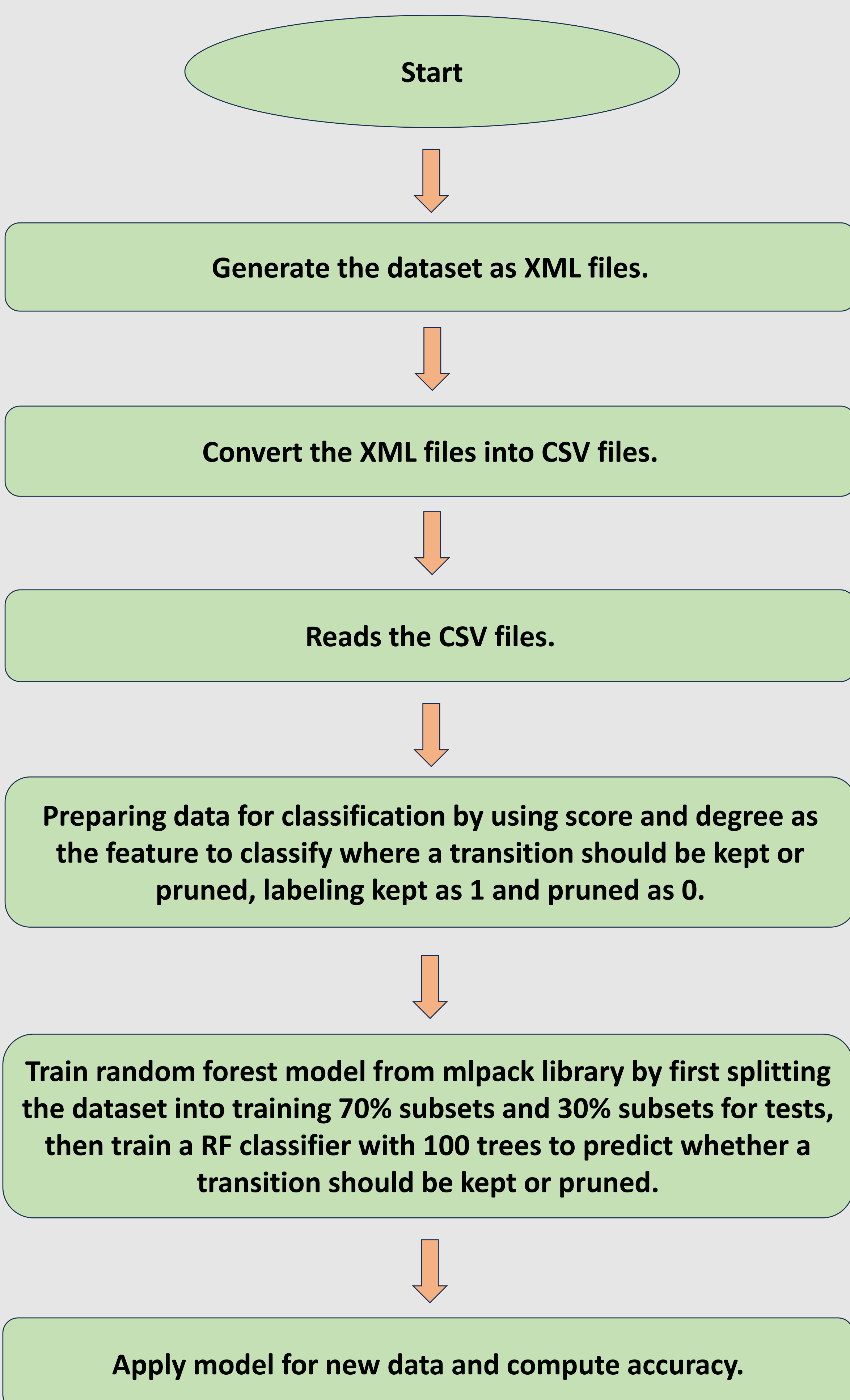
Example

- NAPOLY is a two-dimensional array of processor elements where each processor element consists of RAM and gates.
- Datasets are mapped on the accelerator before starting the operation.
- Dataset represents labels/scores for each edge

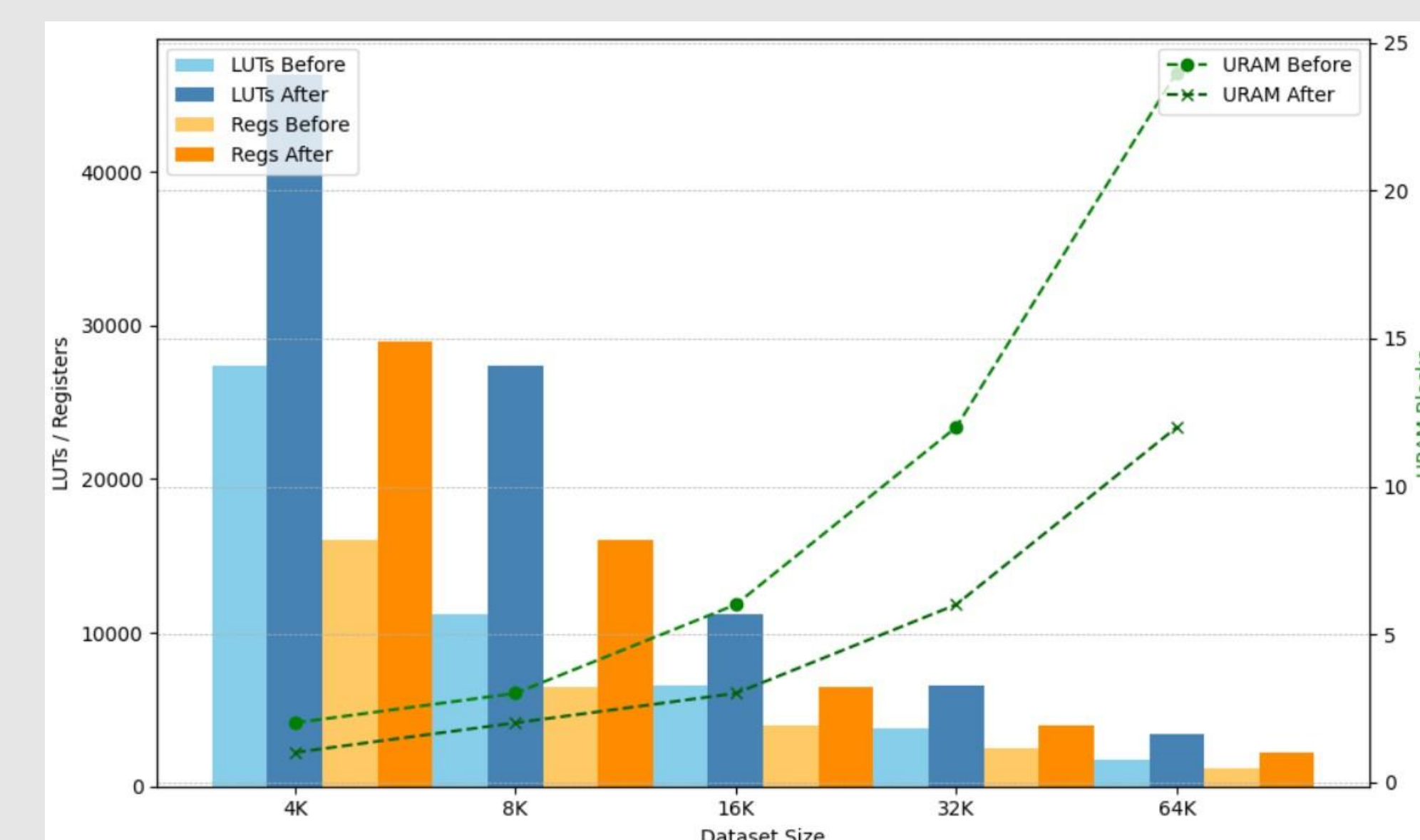


Proposed Work

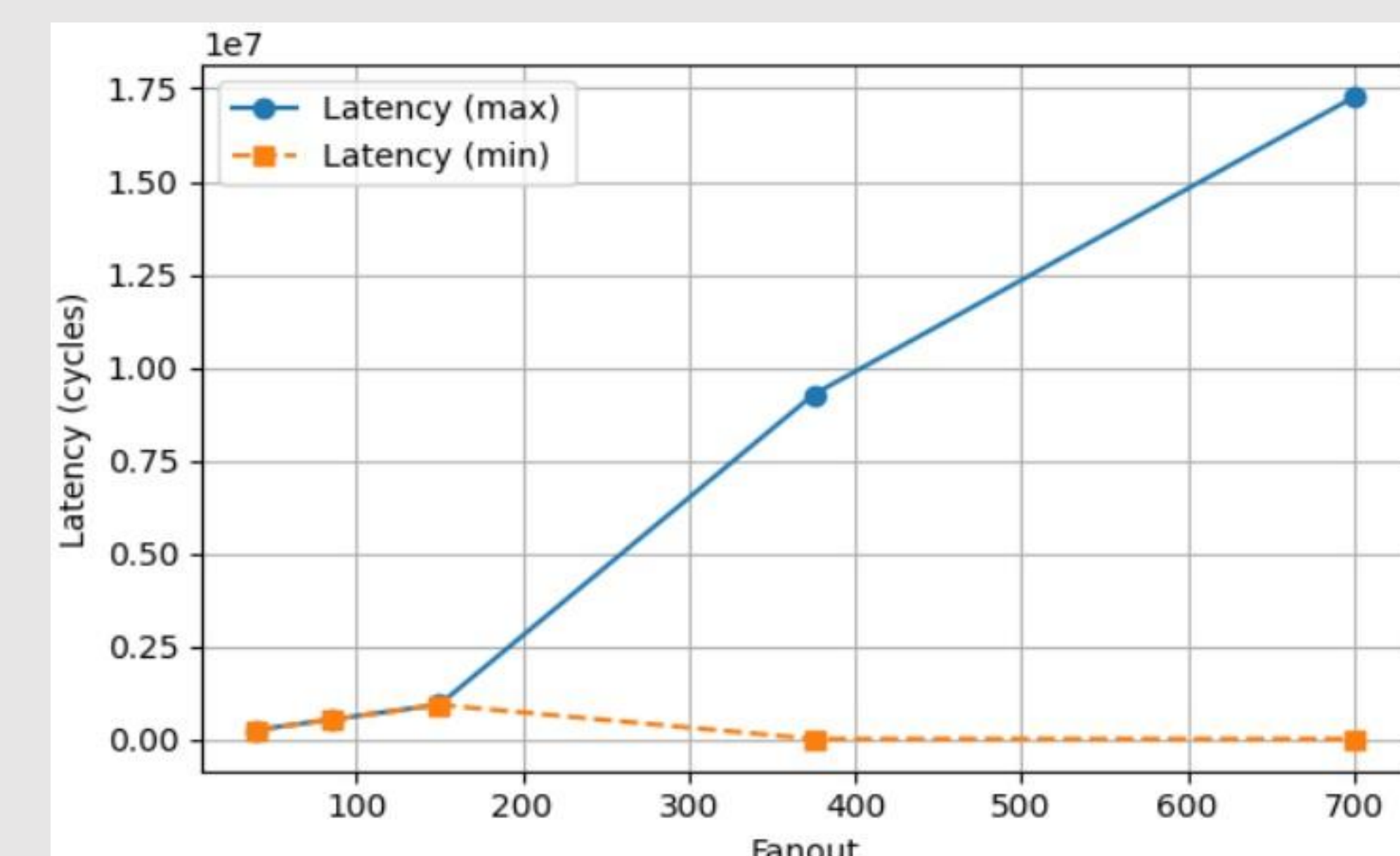
- Develop an intelligent optimization method that eliminates unnecessary states/edges and deduct any malicious information before mapped on processor elements.
- Test and compare the results with state-of-the-art designs.



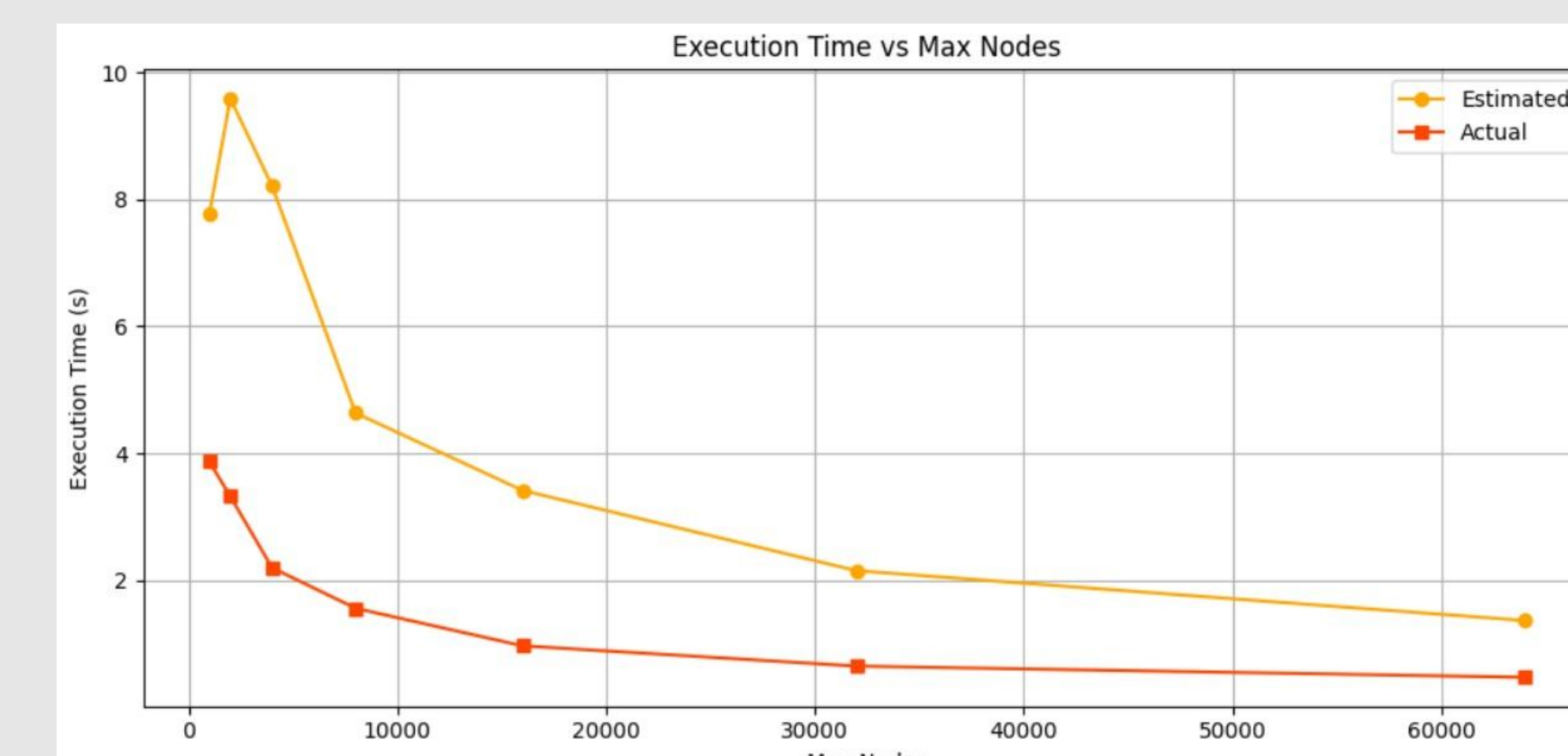
Results



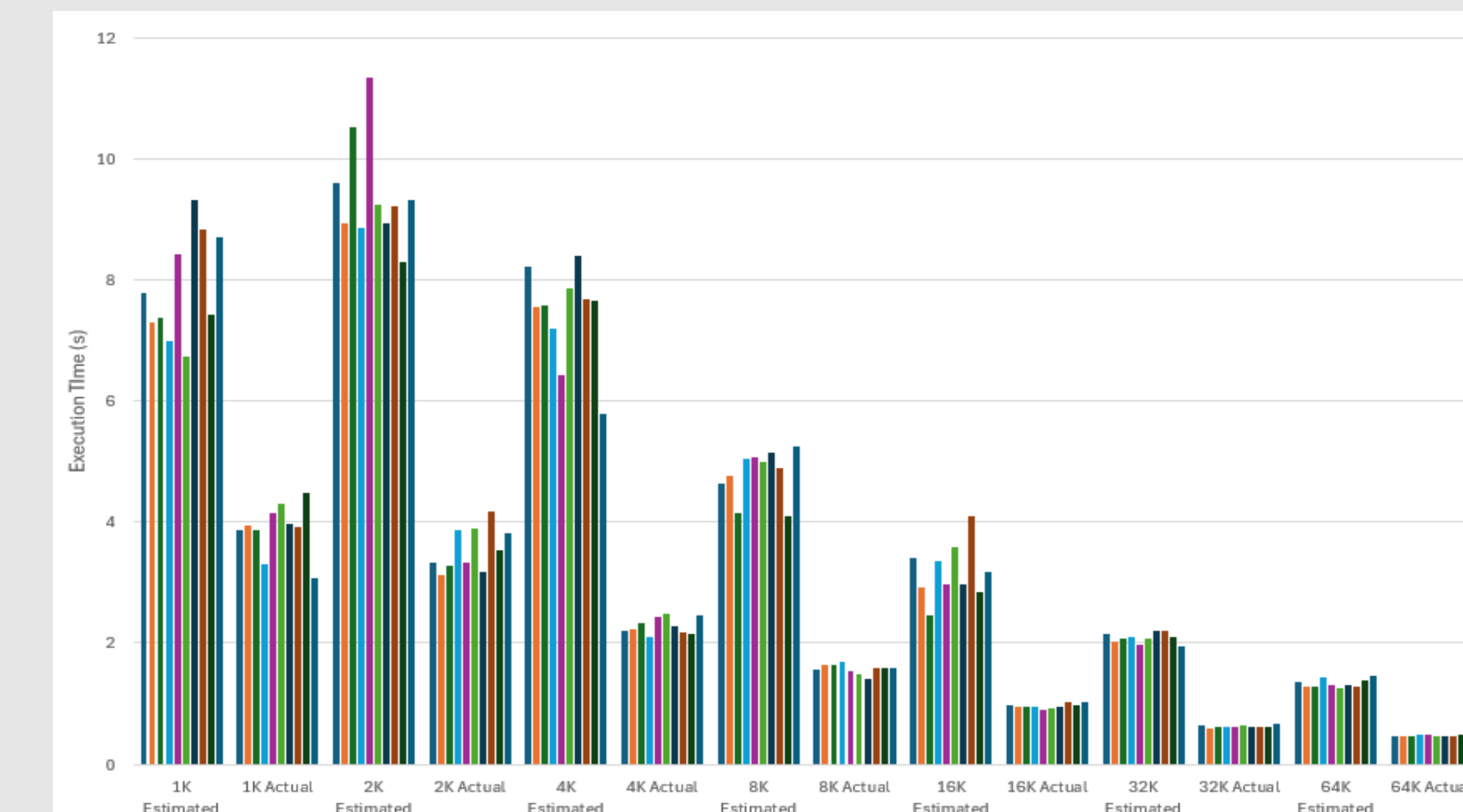
Number of LUTs, Registers, URAM in different size of datasets.



Max and Min number of Latency cycles according to number of Fanout.



Execution Time in secs VS the number of max nodes.



Execution Time in secs VS the different sizes of datasets.

Resources

- Yu, Tiffany, et al. "ML-Based Automata Simplification for Symbolic Accelerators." *arXiv preprint arXiv:2507.08751* (2025).
- Karbowiczak, Ryan, and Rasha Karakchi. "Optimizing Sequence Alignment with Scored NFAs." *arXiv preprint arXiv:2501.02162* (2025). Run on existing tool to eliminate the redundancy

Acknowledgements

This project is accepted for Magellan Scholar, Magellan Apprentice for Capstone Scholar and McNair Fellowship at USC.